NSMO of KSA: Stage 3 Riyadh Finals & Mawhiba IMO Pathway Practice Exercises

1 [NSMO of KSA — Riyadh Finals] A function $f:\mathbb{Z}\to\mathbb{Z}$ satisfies $f(x+y)=f(x)+f(y)+2xy \quad \forall x,y\in\mathbb{Z}$, together with $f(1)=2$ (with additive core $g(x)=f(x)-1x^2$ plotted). Compute $f(7)$.

a. 63

b. 49

c. 56

d. 57

2 [NSMO of KSA — Riyadh Finals] A function $f:\mathbb{Z}\to\mathbb{Z}$ satisfies $f(x+y)=f(x)+f(y)+4xy \quad \forall x,y\in\mathbb{Z}$, together with $f(1)=3$ (with additive core $g(x)=f(x)-2x^2$ plotted). Compute $f(8)$.

a. 152

b. 136

c. 128

d. 120

3 [NSMO of KSA — Riyadh Finals] A function $f:\mathbb{Z}\to\mathbb{Z}$ satisfies $f(x+y)=f(x)+f(y)+6xy \quad \forall x,y\in\mathbb{Z}$, together with $f(1)=4$ (with additive core $g(x)=f(x)-3x^2$ plotted). Compute $f(9)$.

a. 252

b. 279

c. 243

d. 225

4 [NSMO of KSA — Riyadh Finals] A function $f:\mathbb{Z}\to\mathbb{Z}$ satisfies $f(x+y)=f(x)+f(y)+8xy \quad \forall x,y\in\mathbb{Z}$, together with $f(1)=5$ (with additive core $g(x)=f(x)-4x^2$ plotted). Compute $f(10)$.

a. 450

b. 400

c. 370

d. 410

5 [NSMO of KSA — Riyadh Finals] Let $v_3(N)$ denote the exponent of the prime 3 in the prime factorization of N. Using the Lifting The Exponent (LTE) Lemma, compute $v_3(7^{27}-1^{27})$.

a. 3

b. 6

c. 4

d. 5

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6 [NSMO of KSA — Riyadh Finals] Let $v_3(N)$ denote the exponent of the prime 3 in the prime factorization of N. Using the Lifting The Exponent (LTE) Lemma, compute $v_3(10^{18} - 1^{18})$.

a. 3

b. 4

c. 6

d. 5

7 [NSMO of KSA — Riyadh Finals] Let $v_3(N)$ denote the exponent of the prime 3 in the prime factorization of N. Using the Lifting The Exponent (LTE) Lemma, compute $v_3(19^{9} - 1^{9})$.

a. 4

b. 3

c. 6

d. 5

8 [NSMO of KSA — Riyadh Finals] Let $v_3(N)$ denote the exponent of the prime 3 in the prime factorization of N. Using the Lifting The Exponent (LTE) Lemma, compute $v_3(14^{36} - 5^{36})$.

a. 3

b. 6

c. 5

d. 4

9 [NSMO of KSA — Mawhiba IMO Camp] Over all positive real numbers $a, b, c > 0$, find the exact minimum value of the cyclic expression $\frac{2a+1b+1c}{b+c} + \frac{2b+1c+1a}{c+a} + \frac{2c+1a+1b}{a+b}$.

a. 5

b. 9

c. 6

d. $\frac{9}{2}$

10 [NSMO of KSA — Mawhiba IMO Camp] Over all positive real numbers $a, b, c > 0$, find the exact minimum value of the cyclic expression $\frac{2a+2b+2c}{b+c} + \frac{2b+2c+2a}{c+a} + \frac{2c+2a+2b}{a+b}$.

a. 8

b. 9

c. 12

d. $\frac{15}{2}$

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11 [NSMO of KSA — Mawhiba IMO Camp] Over all positive real numbers $a, b, c > 0$, find the exact minimum value of the cyclic expression $\frac{2a+3b+3c}{b+c} + \frac{2b+3c+3a}{c+a} + \frac{2c+3a+3b}{a+b}$.

a. 12

b. 11

c. 15

d. $\frac{21}{2}$

12 [NSMO of KSA — Mawhiba IMO Camp] Over all positive real numbers $a, b, c > 0$, find the exact minimum value of the cyclic expression $\frac{2a+4b+4c}{b+c} + \frac{2b+4c+4a}{c+a} + \frac{2c+4a+4b}{a+b}$.

a. 14

b. 18

c. $\frac{27}{2}$

d. 15

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